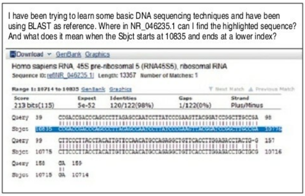
# Bioinformatics: A Shot of Life to Information Technology

*Deepak Halan is associate professor at School of Management Sciences, Apeejay Stya University*

Genomics is an area of study wherein we analyse the DNA sequence of an organism to determine aspects such as genes code behind beneficial traits, genes involved in inherited diseases and much more. This is a dynamic and evolving field and new discoveries are being made every day. Given the speed of progress and the huge amount of data and information being generated, scientists are always looking for better ways to store and analyse this information.

Hence, we have an emerging new area of research that combines the study of biotechnology and the use of computers. This field is referred to as bioinformatics and it involves the use of various IT aspects such as Internet tools, artificial intelligence (AI) and other advanced computational methods to enable storage, retrieval and analysis of data generated from DNA sequencing.

*Fig. 1: As an interdisciplinary field of science, bioinformatics combines computer science, statistics, mathematics and engineering to analyse and interpret biological data (Image courtesy: www.class-central.com)*



I have been trying to learn some basic DNA sequencing techniques and have been using BLAST as reference. Where in NR_046235.1 can I find the highlighted sequence? And what does it mean when the Sbjct starts at 10835 and ends at a lower index?

Homo sapiens RNA, 45S pre-ribosomal 5 (RNA45S5), ribosomal RNA

Sequence ID: ref|NR_046235.1| Length: 13357 Number of Matches: 1

Range 1: 10714 to 10835 GenBank Graphics

| Score | Expect | Identities | Gaps | Strand |
|---|---|---|---|---|
| 213 bits(115) | 5e-52 | 120/122(98%) | 1/122(0%) | Plus/Minus |

```
Query  39     CCGACCGACCCAGCCCTTAGAGCCAATCCTTATCCCGAAGTTACGGATCCGGCTTGCCGA  98
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  10835  CCGACCGACCCAGCCCTTAGAGCCAATCCTTATCCCGAAGTTACGGATCCGGCTTGCCGA  10776

Query  99     CTTCCCTTACCTACATTGTTCCAACATGCCAGAGGCTGTTCACCTTGGAGACCTACTG-G  157
              |||||||||||||||||||||||||||||||||||||||||||||||||||||| ||| |
Sbjct  10775  CTTCCCTTACCTACATTGTTCCAACATGCCAGAGGCTGTTCACCTTGGAGACCTGCTGCG  10716

Query  158    GA  159
              ||
Sbjct  10715  GA  10714
```

*Fig. 2: A query found on Biology Stack Exchange, which is a free question and answer site for biology researchers, academics and students (Image courtesy: biology.stackexchange.com/questions/8160/blast-dna-sequences-reversed)*

## Application areas

Bioinformatics is the combination of biology, computer science and IT into a single stream. The ultimate objective is to seek new biological insights and develop a global outlook. Bioinformatics incorporates three vital sub-disciplines. These are:

1. Development of new algorithms and statistics to evaluate relationships between members of large data sets

2. Analysis and understanding of different types of data such as nucleotide and amino acid sequences, protein domains and protein structures

3. Creation and application of tools that enable efficient access and management of diverse types of information

***International exchange of insights among scientists.*** Scientists around the world need to pool in information on genome research. The Internet connects millions of individual networks across the globe via email, news groups, chat and other media. The information that is shared can be in the form of text, graphics, audio, video and animation.

The Internet offers several useful user-friendly tools for scientists, such as one that is used to analyse DNA and protein sequences. Evolution of more advanced IT tools and increased sharing of information will lead to higher efficiency in genome research and hence superior methods to diagnose diseases, identify beneficial traits and offer treatment for crop, animal and human diseases.

***Information storage and analysis.*** In simple terms, DNA is a molecule made from sugar, phosphate and bases called guanine (G), cytosine (C), adenine (A) and thymine (T). The different combinations of these bases form the DNA in plants, animals, bacteria, yeast and fungi. An infinite number of combinations of these bases are possible, for example, AAGCT, CCAGT, TACGGT and many more. With the help of databases and supercomputers, scientists are trying to determine the entire DNA sequence of the numerous living beings.

Human Genome Project is one of the largest initiatives. Findings generated from DNA sequencing will help identify genes (units of DNA that code for specific traits),